Linearized Commutativity Geometry on the Generator-Set Moduli Space

Full-Matrix Jacobian Certificates, Normality Gates, and Typed Spectral Registrations

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Abstract

Problem. Weighted generator averages define a commutativity locus in the generator-set moduli space, but commutativity alone does not provide orthogonal joint-spectral projectors. Normality and coherent projector continuation are additional gates. A single untyped accessibility ladder would conflate operator products, words, and Lie brackets, which are distinct mathematical objects.

Approach. We split the paper into two layers. First, we study the commutator map

$$C_{\mathrm{comm}}(w) = [Q_T(w), H_T(w)]$$

at the canonical Rubik point. Every Jacobian matrix is encoded by all real and imaginary entries; no skew-Hermitian assumption is imposed. We then adjoin the linearized normality equations for Q_T and H_T. Second, only after commutativity, Hermiticity, normality, and projector checks pass do we register joint sectors and compare the typed supports $R_1^{\mathrm{op}} := R_1[\rho(S)]$ and $R_1^{\mathrm{Lie}} := R_1[X]$.

Results. The full complex-real commutator Jacobian has numerical rank 11 and nullity 7. Its nonzero singular values occur in groups

$$0.585314^{\times 3}, \quad 0.532870^{\times 3}, \quad 0.346944^{\times 2}, \quad 0.200308^{\times 3}.$$

The combined commutativity-normality derivative has numerical rank 14 and nullity 4. Its kernel is spanned numerically by uniform half-turn scaling and three inverse-pair-symmetric quarter-turn axis directions. This kernel contains both exact class-scaling gauges, while the three QT-axis directions have sample points that pass the declared numerical commutativity and normality gates. At the canonical point, the validated joint decomposition has 9 sectors and typed support counts 438 and 408. At quarter-turn axis weight 1.1, each tested axis has 15 sectors; $R_1^{\mathrm{op}} = 1006$, while the declared finite-order logarithm branch gives $R_1^{\mathrm{Lie}} = 832$ on each of the three axes.

Boundary. Rank 11 is a linearized commutativity-kernel certificate, not a proof of a seven-dimensional smooth zero-set manifold. The pointwise sector-count change from 9 to 15 is a normality-gated computational record, not a refinement or global wall theorem. Routed composition, full words, commutators, word depth, and Lie depth require separate fields and separate certificates. No moving-wall theorem is claimed.

Notation and Claim Layers

The paper uses four claim-status levels.

1. **Theorem:** finite-dimensional identities derived from the declared formulas, including lemmas and propositions under stated hypotheses.

2. **Computational Certificate:** matrix ranks, singular values, kernel directions, residuals, and projector checks with declared dtype and tolerance.

3. **Computational Observation:** pointwise sector-count and typed-support records without coherent continuation.

4. **Research Program:** statements explicitly identified as open and not used in the proved or certified results.

The domain gates are

$$\mathcal{M}_{>0} \supseteq \Sigma_{\text{comm}} \supseteq \Sigma_{\text{normal}} \supseteq \Sigma_{\text{spec}}^{(\nu)}.$$

Here $\Sigma_{\text{spec}}^{(\nu)}$ denotes one normal spectral chart in an atlas. No global coherent labeling of joint sectors is assumed.

The accessibility objects are branched:

$$R_1^{\text{op}} \longrightarrow C_d^{\text{op}},\ W_d^{\text{op}} \longrightarrow D_{\text{route}}^{\text{op}},\ D_{\text{word}}^{\text{op}},$$

and

$$R_1^{\text{Lie}} \longrightarrow R_2^{\text{Lie}} \longrightarrow D_{\text{Lie}}.$$

No arrow between the two branches is asserted without an explicit bridge theorem. These arrows denote construction and audit order only; they do not encode functional determination, logical implication, or completion.

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1 Introduction

For a fixed finite joint arrangement P, one may vary only the linear functional [6]. Here a different question is considered: what remains valid when the generator weights vary and the pair itself becomes

$$(Q_T, H_T) = (Q_T(w), H_T(w))?$$

Three logically distinct problems appear.

First, the two weighted averages may fail to commute. The global algebraic object is therefore the commutativity locus Σ_{comm}.

Second, even a commuting pair need not be normal. Orthogonal joint sectors are available only after a normality gate. Even after that gate, smooth projector fields are local chart data; collisions, multiplicity changes, and monodromy can prevent a single global coherent labeling.

Third, once sectors have been registered, several typed accessibility notions can be computed. A support graph of represented group elements, a routed projected product, a full word, a projected commutator, and a Lie-depth certificate are distinct typed objects and are not interchangeable. Papers III and V establish the corresponding static separations [7, 5]. The moving theory must preserve those types.

The contributions are deliberately narrower than a universal wall theory:

1. an exact Jacobian formula for the normalized commutator map;

2. a corrected full-matrix rank-11 computational certificate;

3. a linearized normality gate with rank 14 and nullity 4;

4. two exact class-scaling gauge families and three numerically certified QT-axis sample points;
5. pointwise projector and typed-support registrations at those points;
6. a conditional object language for future moving accessibility audits.

The paper does not claim that the commutativity zero set is a smooth seven-dimensional manifold. It does not claim a complete $\Sigma_{R_1}/\Sigma_{R_2}/\Sigma_D$ hierarchy. It does not identify graph reachability with matrix composition or Lie depth. The pointwise registrations can serve as endpoints for a moving analysis only after coherent projector matching and chart regularity are certified.

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2 Weighted Generator Moduli

Let $S = \{g_1, \dots, g_m\}$ be a fixed finite generator family and let

$$\rho(g_i) \in U(V)$$

be its declared finite-dimensional unitary realization. For the local differential analysis, the generator-weight moduli space is the open positive cone

$$\mathcal{M}_{>0} = (0, \infty)^m.$$

The larger nonnegative domain

$$\mathcal{M}_+ = \left\{ w \in \mathbb{R}^m_{\geq 0} : \sum_{i \in \mathrm{QT}} w_i > 0,\ \sum_{i \in \mathrm{HT}} w_i > 0 \right\}$$

is available for boundary probes. The canonical point $w_0 = \mathbf{1}$ is an interior point of $\mathcal{M}_{>0}$, and the tested weights $1 + \varepsilon$ remain in this domain for $\varepsilon > -1$.

For the canonical Rubik realization, $m = 18$ and $V \cong \mathbb{C}^{228}$. The generators split into twelve quarter turns and six half turns.

2.1 Weighted QT/HT averages

On the open domain where both class totals are nonzero, define

$$Q_T(w) = \frac{\sum_{i \in \mathrm{QT}} w_i \rho(g_i)}{\sum_{i \in \mathrm{QT}} w_i}, \qquad H_T(w) = \frac{\sum_{i \in \mathrm{HT}} w_i \rho(g_i)}{\sum_{i \in \mathrm{HT}} w_i}.$$

The normalized pair is invariant under independent positive class scalings:

$$(a, b) \cdot w_i = \begin{cases} a w_i, & i \in \mathrm{QT}, \\ b w_i, & i \in \mathrm{HT}, \end{cases} \qquad a, b > 0.$$

Thus its effective parameter space carries the gauge quotient $\mathcal{M}_{>0}/((\mathbb{R}_{>0})_{\mathrm{QT}} \times (\mathbb{R}_{>0})_{\mathrm{HT}})$. We retain redundant positive coordinates for the Jacobian calculation. This quotient applies to the normalized QT/HT pair; the global average $A(w)$ below is not invariant under independent class scalings. Clearing denominators produces polynomial commutativity equations, but doing so may add excluded boundary components if the positive-total restriction is forgotten.

The global weighted average

$$A(w) = \frac{\sum_i w_i \rho(g_i)}{\sum_i w_i}$$

has a spectrum wherever the total weight is nonzero. Its definition does not require QT/HT commutativity. The $A(w)$-spectrum and the QT/HT collision quotient are therefore different observables with different domains.

2.2 Commutativity and normality loci

Define

$$\Sigma_{\mathrm{comm}} = \{w \in \mathcal{M}_{>0} : [Q_T(w), H_T(w)] = 0\}.$$

For a matrix M, write

$$N(M) = MM^* - M^*M.$$

The normal commuting locus is

$$\Sigma_{\mathrm{normal}} = \{w \in \Sigma_{\mathrm{comm}} : N(Q_T(w)) = N(H_T(w)) = 0\}.$$

Inverse-pair-symmetric weight families are Hermitian and hence normal. Generic independent weight changes need not preserve this property.

2.3 Normal spectral charts

A normal spectral chart $\Sigma^{(\nu)}_{\mathrm{spec}}$ is a relatively open local piece of Σ_{normal} on which the joint multiplicity pattern is fixed and orthogonal joint projectors can be labeled coherently:

$$I = \sum_{i \in I_\nu} Q_i(w), \qquad Q_i(w)Q_j(w) = \delta_{ij} Q_i(w), \qquad Q_i(w)^* = Q_i(w).$$

A moving generator family may require an atlas $\{\Sigma^{(\nu)}_{\mathrm{spec}}\}_\nu$. Pointwise simultaneous diagonalization does not by itself prove coherent projector continuation on a neighborhood. Standard perturbation theory supplies the relevant background for isolated spectral clusters [16].

On one such chart the joint spectrum is a finite weighted arrangement

$$P(w) = \{(q_i(w), h_i(w), d_i(w))\}_{i \in I_\nu}.$$

For fixed α, the collision quotient uses

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h.$$

The fixed-arrangement theorem applies pointwise after $P(w)$ has been registered; it does not prove that the arrangement or its projectors vary smoothly with w [6].

3 Full-Matrix Linearized Commutativity Certificate

Let

$$w_0 = \mathbf{1} = (1, \dots, 1).$$

At this point the canonical averages satisfy

$$\|[Q_T(w_0), H_T(w_0)]\|_F = 3.766 \times 10^{-16}$$

in the declared `complex128` realization.

3.1 Exact derivative formula

Lemma 3.1 (Normalized commutator derivative)**.**
At $w_0 = \mathbf{1}$, without assuming exact QT/HT commutation, the derivative with respect to one QT weight is

$$J_k = \frac{[\rho(g_k), H_T(w_0)] - [Q_T(w_0), H_T(w_0)]}{12}, \qquad k \in \mathrm{QT},$$

and the derivative with respect to one HT weight is

$$J_k = \frac{[Q_T(w_0), \rho(g_k)] - [Q_T(w_0), H_T(w_0)]}{6}, \qquad k \in \mathrm{HT}.$$

Proof. *For a normalized weighted average*

$$M(w) = \frac{\sum_i w_i M_i}{\sum_i w_i},$$

the derivative at equal unit weights is

$$\frac{\partial M}{\partial w_k} = \frac{M_k - M(w_0)}{|S|}.$$

Differentiating $[Q_T, H_T]$ gives the displayed formulas directly. If exact commutation is separately certified at w_0, the final commutator term in each numerator vanishes and the formulas simplify to the shorter conditional expressions.

3.2 Full complex-real encoding

The QT derivatives J_k are not individually skew-Hermitian. Their maximum skew-Hermitian residual is

$$\|J_k + J_k^*\|_F = 0.200308.$$

Consequently, retaining only a strict upper triangle is not a valid ambient encoding. We therefore define the full realification

$$\mathfrak{R}(M) = \begin{pmatrix} \operatorname{vec} \mathfrak{R}M \\ \operatorname{vec} \mathfrak{I}M \end{pmatrix} \in \mathbb{R}^{2N^2}.$$

The computational Jacobian is the $2(228)^2 \times 18 = 103968 \times 18$ real matrix

$$\mathcal{J}_{\mathrm{comm}} = [\mathfrak{R}(J_1) \ \cdots \ \mathfrak{R}(J_{18})].$$

The computation uses the unconditional formulas in Lemma 3.1, including the machine-scale baseline commutator terms.

Computational Proposition 3.2 (Linearized commutativity-kernel certificate). In the declared realization and at relative singular-value threshold 10^{-10},

$$\operatorname{rank} \mathcal{J}_{\mathrm{comm}} = 11, \qquad \dim \ker \mathcal{J}_{\mathrm{comm}} = 7.$$

The singular-value groups are

singular value	multiplicity
0.585314	3
0.532870	3
0.346944	2
0.200308	3
numerical zero	7

The rank boundary is numerically separated by

$$\sigma_{11} = 2.003084041924 \times 10^{-1}, \qquad \sigma_{12} = 2.415461205698 \times 10^{-15},$$

with retained/discarded ratio 8.293×10^{13}.

The twelve QT Jacobians have Frobenius norm 0.309320 and the six HT Jacobians have norm 0.426730.

The kernel contains uniform QT scaling, uniform HT scaling, and three quarter-turn axis-symmetric directions. Single-coordinate deletions are not in the kernel. These are statements about the derivative. They do not prove that every kernel vector integrates to a curve in the exact zero set.

3.3 What rank 11 does not prove

The space

$$\ker DC_{\mathrm{comm}}|_{w_0}$$

is the kernel of the linearized commutator constraint at a numerically registered near-zero point. Because exact commutation at w_0 has not been certified here, this kernel is not unconditionally identified with the Zariski tangent space of Σ_{comm}. If exact membership is later established, promoting this kernel to the tangent space of a smooth seven-dimensional local manifold still requires a constant-rank or implicit-function argument on the actual zero set. The reported finite samples and random searches do not supply that proof. Smooth-manifold and local algebraic-geometry language is therefore used only conditionally [17, 8, 12].

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4 The Normality Gate

Commutativity is necessary but insufficient to guarantee the orthogonal sectorization used later. Therefore, we linearize the combined map

$$F(w) = (C_{\mathrm{comm}}(w), N(Q_T(w)), N(H_T(w))).$$

For an operator M and variation $\dot{M}$,

$$DN_M(\dot{M}) = \dot{M}M^* + M\dot{M}^* - \dot{M}^*M - M^*\dot{M}.$$

Every component is again encoded by full real and imaginary entries.

Computational Proposition 4.1 (Combined linearized constraint kernel). At the numerically registered point $w_0 = \mathbf{1}$, the combined derivative has

$$\operatorname{rank} DF|_{w_0} = 14, \qquad \dim\ker DF|_{w_0} = 4.$$

Its singular-value groups are

$$0.585314^{\times 3}, \quad 0.532870^{\times 3}, \quad 0.482726^{\times 2}, \quad 0.346944^{\times 2},$$

$$0.283279^{\times 1}, \quad 0.200308^{\times 3}, \quad 0^{\times 4}.$$

The combined-map rank boundary is

$$\sigma_{14} = 2.003084041924 \times 10^{-1}, \qquad \sigma_{15} = 2.238468379596 \times 10^{-15},$$

with retained/discarded ratio 8.948×10^{13}.

Four interpretable vectors span the numerical kernel:

1. uniform scaling of all six HT weights;
2. equal variation of the four QT weights on axis 0;
3. equal variation of the four QT weights on axis 1;
4. equal variation of the four QT weights on axis 2.

The sum of the three QT-axis vectors is uniform QT scaling, the second exact class-scaling gauge direction.

Their projection residuals onto the numerical kernel are between 1.0×10^{-15} and 5.8×10^{-15}.

Because the baseline commutator is registered only to numerical precision, this kernel is not identified unconditionally with $T_{w_0}\Sigma_{\text{normal}}$.

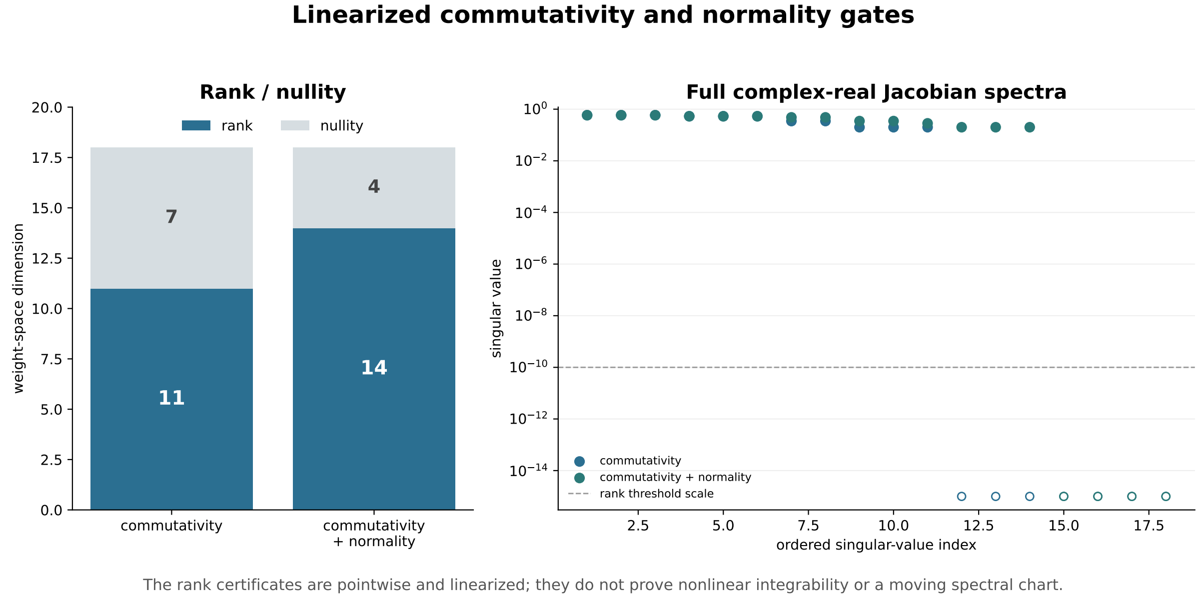

Figure 1. Full complex-real Jacobian certificates at the canonical point. Adding the linearized normality equations changes the registered rank/nullity from 11/7 to 14/4, with a clear retained/discarded singular-value gap. These are pointwise linearized certificates, not a nonlinear integrability or moving-chart theorem.

4.1 Exact gauge families and tested sample points

The four kernel vectors do not have the same certification status.

Uniform HT and uniform QT scaling are exact gauges of the normalized pair. Multiplying all weights in either class by the same positive factor leaves the corresponding normalized average unchanged. At $\varepsilon = 0.1$, the measured operator drifts are 1.526×10^{-15} for HT scaling and 5.230×10^{-16} for QT scaling.

For each axis $a \in \{0, 1, 2\}$, set

$$w_i(\varepsilon) = \begin{cases} 1+\varepsilon, & g_i \text{ is a QT on axis } a, \\ 1, & \text{otherwise.} \end{cases}$$

The inverse-pair weights remain equal, so $Q_T(w)$ and $H_T(w)$ are Hermitian. At $\varepsilon = 0.1$, all three sampled points satisfy commutator and normality residuals below 2×10^{-16}.

This gives two exact class-scaling gauge families and three certified QT-axis sample points. No interval-wide commutativity sweep or exact commutation proof is claimed for the QT-axis parameterizations. The samples do not classify the nonlinear set Σ_{normal}.

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5 Normality-Gated Spectral Registrations

The sector algorithm is called only after the pair certificate passes:

$$\|[Q_T, H_T]\|_F, \quad \|N(Q_T)\|_F, \quad \|N(H_T)\|_F, \quad \|Q_T - Q_T^*\|_F, \quad \|H_T - H_T^*\|_F < 10^{-8}.$$

For the resulting projectors, the audit checks

$$\max_i \|Q_i^2 - Q_i\|_F, \qquad \max_i \|Q_i - Q_i^*\|_F, qquad \max_{i \neq j} \|Q_i Q_j\|_F, qquad \left\| \sum_i Q_i - I \right\|_F.$$

The maximum projector residual in the registered samples is 2.390×10^{-14}.

5.1 Typed support conventions

For represented group elements define

$$R^{\text{op}}_{1,g}(i,j) = \mathbf{1}[Q_i \rho(g) Q_j \neq 0].$$

For declared skew-Hermitian logarithms define

$$R^{\text{Lie}}_{1,g}(i,j) = \mathbf{1}[Q_i X_g Q_j \neq 0].$$

The direction convention is always

$$Q_i Y Q_j \neq 0 \quad \Longleftrightarrow \quad j \longrightarrow i.$$

The reported count is the number of supported labeled blocks (g, i, j), including diagonal blocks. It is not an unlabelled simple-graph edge count.

For this audit, X_g is the finite-order principal logarithm determined by the order-2 or order-4 spectral projectors, with arguments in $(-\pi, \pi]$ and $\log(-1) = i\pi$. This branch declaration is part of the realization. A different logarithm branch can change R_1^{Lie} without changing R_1^{op}.

5.2 Registered samples

All computations use `complex128`, support and clustering threshold 10^{-8}, and the same 18-generator realization.

Sample	sectors	R_1^{op}	R_1^{Lie}
canonical $w = \mathbf{1}$	9	438	408
QT axis 0, $\varepsilon = 0.1$	15	1006	832
QT axis 1, $\varepsilon = 0.1$	15	1006	832
QT axis 2, $\varepsilon = 0.1$	15	1006	832

The sorted sector-dimension censuses are

$$\begin{aligned}\mathrm{dims}_{\mathrm{canonical}} &= (1, 2, 8, 20, 26, 27, 39, 39, 66),\\ \mathrm{dims}_{\mathrm{axis}} &= (1, 1, 1, 8, 9, 13, 13, 13, 13,\\ &\qquad 18, 20, 22, 26, 26, 44).\end{aligned}$$

Thus, the pointwise registrations exhibit a sector-count change from 9 to 15 alongside a change in both typed direct-support counts. The operator and Lie direct-support counts are not equal. The canonical Lie-support gap has smallest retained block norm 0.5236 and largest discarded block norm 1.83×10^{-14}; across the three axis samples the smallest retained norm is 3.53×10^{-3} and the largest discarded norm is 1.50×10^{-13}. The logarithm branch, dtype, and threshold therefore remain part of the declared numerical registration.

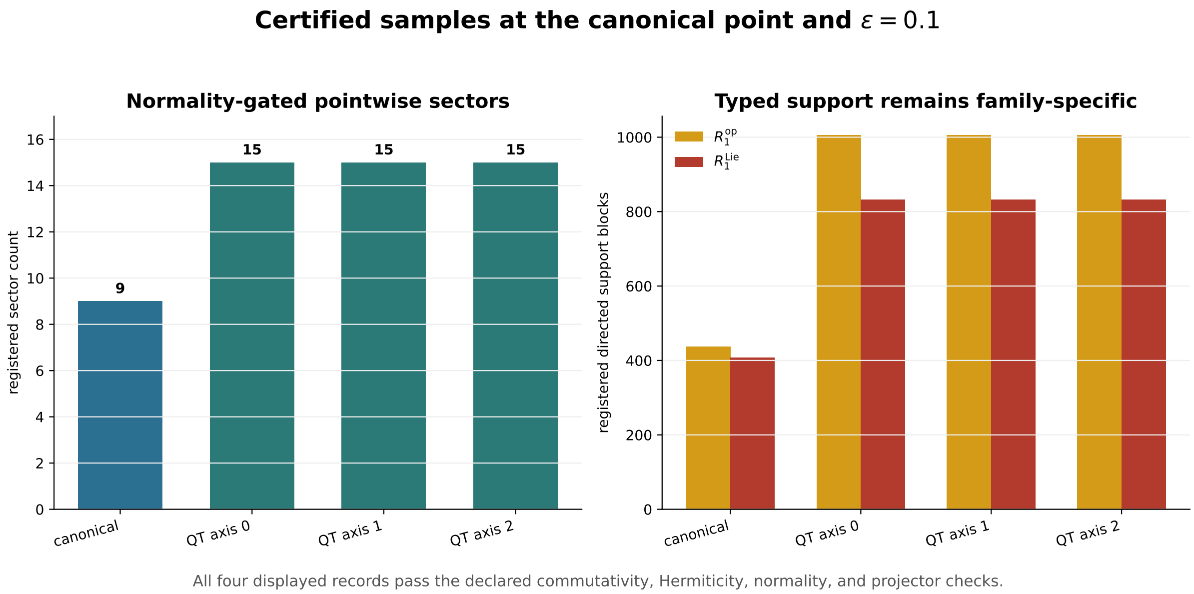

Figure 2. Normality-gated pointwise registrations. The canonical point has nine sectors, while each of the three tested QT-axis samples has fifteen; the operator-family and Lie-family direct-support counts are displayed separately. All four records pass the declared numerical gates, but the figure does not supply projector continuation along an intervening path.

These samples do not yet prove a wall location or a smooth projector chart connecting the endpoints. Such a claim requires coordinate-matched projector continuation and threshold-stability sweeps along the intervening path.

6 Conditional Typed Accessibility Fields

This section defines the typed objects conditionally on a normal spectral chart $U = \Sigma^{(\nu)}_{\mathrm{spec}}$ with declared smooth projectors $Q_i(w)$.

6.1 Operator and word branch

For $Y_g = \rho(g)$, define direct blocks

$$B^{\mathrm{op},g}_{ij}(w) = Q_i(w) Y_g Q_j(w).$$

For an intermediate sequence $i_0 = j, i_1, \ldots, i_d = i$, define the routed product

$$C^{\mathrm{op},g_d\cdots g_1}_{i_d\cdots i_0}(w) = Q_{i_d} Y_{g_d} Q_{i_{d-1}} \cdots Q_{i_1} Y_{g_1} Q_{i_0}.$$

The full word block is

$$W^{\mathrm{op},g_d\cdots g_1}_{ij}(w) = Q_i Y_{g_d} \cdots Y_{g_1} Q_j.$$

Explicit summation over intermediate sectors demonstrates the separation between these objects:

$$W^{\mathrm{op},g_d\cdots g_1}_{ij} = \sum_{i_1,\ldots,i_{d-1}} C^{\mathrm{op},g_d\cdots g_1}_{i\,i_{d-1}\cdots i_1 j}.$$

Different routed terms may cancel. Define routed-composition depth and word depth separately, with $\inf \emptyset = \infty$ only for exact saturated objects. Numerical cutoffs use `unreached`, never infinity.

6.2 Lie branch

For declared skew-Hermitian X_g, define

$$B^{\mathrm{Lie},g}_{ij}(w) = Q_i(w) X_g Q_j(w),$$

and

$$C^{\mathrm{Lie},g,h}_{ij}(w) = Q_i(w)[X_g, X_h] Q_j(w).$$

Their support shadows are R_1^{Lie} and R_2^{Lie}. Higher Lie depth D_{Lie} is defined from a declared Hall filtration. Exact infinity requires a full Lie-closure certificate; a finite computation reports cutoff-unreached channels. Hall bases and geometric control supply the standard background [11, 18, 15, 2].

R_2^{Lie} is commutator support, not a generic repair theorem. Static cancellation and image–kernel incidence are mechanisms, not moving-wall labels.

6.3 Conditional local constancy

Proposition 6.1 (Finite typed-shadow constancy)**.**

Fix a normal spectral chart with continuous projector fields and fix a finite declared list of block, routed-product, word, commutator, or Hall matrices. On any open region where the ranks of all matrices in that list remain constant, their finite Boolean support shadows are locally constant.

Proof. *A matrix has zero support exactly when its rank is zero. Constant rank therefore fixes every declared zero/nonzero status.*

This proposition is finite and typed. It does not say that R_1^{Lie} determines R_2^{Lie}, that support paths determine routed products, or that a finite cutoff determines exact depth.

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7 Claim Status and Boundary

The table uses the four claim levels. Lemma and exact identity are result types inside the Theorem level; negative and unestablished implications are listed separately as boundaries.

Claim	Status
derivative formulas for the normalized QT/HT commutator	Theorem
full complex commutator Jacobian has rank 11, nullity 7	Computational Certificate
combined commutativity-normality derivative has rank 14, nullity 4	Computational Certificate
four displayed directions span the numerical combined kernel	Computational Certificate
uniform QT/HT gauges leave the normalized pair fixed	Theorem
three QT-axis points pass numerical commutator, Hermiticity, and normality audits	Computational Certificate
canonical and axis-sample projectors are orthogonal and complete	Computational Certificate
registered sector-count change from 9 to 15	Computational Observation
typed R_1^{op} and R_1^{Lie} counts	Computational Observation
coherent projector continuation between registered endpoints	Research Program
moving routed/word/Lie depth walls	Research Program

The boundary is explicit: the rank-11 kernel is not a proved seven-dimensional smooth Σ_{comm} manifold, and no universal inclusion hierarchy among accessibility walls is claimed.

7.1 Earlier-version correction

An earlier fragmentation calculation incorrectly diagonalized commuting but nonnormal QT averages using a Hermitian eigensolver. The resulting $9 \to 24$–35 sector counts and $438 \to 6334$ untyped support count are invalid because the normality gate was not satisfied, and they are therefore not used here.

Likewise, statements that fragmentation automatically changes the untyped R_2 or D are withdrawn. No corresponding Rubik typed product, commutator, or closure certificate was computed.

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8 Research Program

The open problems below are ordered by mathematical dependency.

8.1 Integrability of the commutativity kernel

Determine local defining equations for Σ_{comm}, first certify whether w_0 belongs to the exact commutativity zero set, and then decide which of the seven linearized directions integrate to exact curves. Rank stability sampled away from the zero set is not a substitute for an implicit-function or local-algebraic certificate.

8.2 Normal spectral atlas

Determine whether the four-dimensional combined linearized constraint kernel extends to actual local charts. Required checks include commutativity and normality along the full path, constant joint multiplicities away from registered collision points, coordinate-matched projector overlaps, principal angles, and monodromy around degeneracies.

8.3 Typed moving fields

On certified charts, compute the branches independently:

$$(R_1^{\mathrm{op}}, C_d^{\mathrm{op}}, W_d^{\mathrm{op}}, D_{\mathrm{route}}^{\mathrm{op}}, D_{\mathrm{word}}^{\mathrm{op}})$$

and

$$(R_1^{\mathrm{Lie}}, R_2^{\mathrm{Lie}}, D_{\mathrm{Lie}}).$$

Only actual matrix products support composition claims. Only projected commutators and a saturated Lie filtration support Lie-depth claims.

Only after a deformation path or chart and its typed matrix fields have been certified may one define corresponding rank/support discriminants. Candidate loci include

$$\Sigma_{R_1^{\mathrm{op}}}, \quad \Sigma_{D_{\mathrm{route}}^{\mathrm{op}}}, \quad \Sigma_{D_{\mathrm{word}}^{\mathrm{op}}}, \quad \Sigma_{R_1^{\mathrm{Lie}}}, \quad \Sigma_{R_2^{\mathrm{Lie}}}, \quad \Sigma_{D_{\mathrm{Lie}}}.$$

These loci need not coincide or form one total inclusion chain. First-order jets may be useful diagnostics, but need not detect every higher-order support change.

8.4 Spectral collision and field loci

The moving layer and field loci Σ_L and Σ_{field} remain candidate spectral stratifications inside certified spectral charts. Their definitions require exact or explicitly registered algebraic coordinates. No universal inclusion with typed accessibility loci is asserted.

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9 Related Work and Novelty Boundary

Finite-group representations and the Rubik realization use standard representation-theoretic background [19, 14]. Commuting normal matrices and isolated spectral-cluster continuation are classical finite-dimensional perturbation theory [16]. Matrix rank and singular-value certificates use standard matrix analysis [13].

The algebraic geometry of commuting matrix pairs is classical [9]. The present parameter family is a structured pullback of a commuting locus with additional normalization and normality constraints; no statement about the global components or irreducibility of that pullback is made here. Numerical simultaneous diagonalization and the sensitivity of joint diagonalizers provide a second direct comparison class [4, 1]. Those works concern diagonalization algorithms and perturbation sensitivity, whereas the present certificate records linearized constraints and pointwise registered spectral data without asserting a continuation theorem.

The local zero-set discussion is adjacent to smooth constant-rank theory and local algebraic geometry [17, 8, 12]. The present paper does not introduce a general perturbation theorem for commuting normal tuples. Its contribution is the corrected full-matrix certificate, the explicit normality gate, and the typed separation required before moving accessibility claims can be formulated.

Association schemes and Bose–Mesner algebras provide neighboring examples of commutative semisimple matrix algebras [3, 10]. They are comparison classes, not asserted identifications of the QT/HT algebra.

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10 Conclusion

The generator-set moduli problem has two distinct layers.

The first is commutativity geometry. At the canonical point, the corrected full complex-real Jacobian has rank 11 and nullity 7. This is a linearized certificate, not a smooth-manifold theorem.

The second is the construction of moving spectral and accessibility fields. That construction begins only after normality and projector continuation have been certified. The combined linearized commutativity-normality map has rank 14 and nullity 4. The kernel contains both exact class-scaling gauges, and the three displayed QT-axis directions have certified sample points. The axis points have 15 registered sectors versus 9 at the canonical point; no projector refinement relation is asserted. Operator and Lie direct-support counts remain distinct.

The correct future architecture is therefore

$$\text{commutativity} \longrightarrow \text{normality} \longrightarrow \text{spectral charts} \longrightarrow \text{typed matrix fields} \longrightarrow \text{typed wall audits.}$$

Skipping any gate in this sequence makes the subsequent accessibility statement undefined or ambiguous. The results therefore establish pointwise typed registrations, not a moving-wall theorem.

Appendix A Computational Artifacts

The following repository artifacts support the linearized certificates and pointwise registrations. The default directory is `experiments/paper6/`; paths are relative to that directory.

Artifact	Role	Short path
A1	full complex-real commutator Jacobian certificate	`validation/tangent_commutator_map.py`
A2	normality gate and pointwise typed registration audit	`validation/normal_spectral_chart_audit.py`
A3	contextual ambient-moduli probes	`validation/generator_moduli_space.py`
A4	tangent-map run summary	`results/_paper6_tangent_commutator_map.txt`
A5	ambient-moduli run summary	`results/_paper6_bifurcation_log.txt`

From the repository root, run an executable artifact as `python experiments/paper6/<short path>`. The numerical reports declare matrix dtype and norm, absolute and optional relative thresholds, reference scale, threshold sweep, projector registration, and any logarithm branch, cutoff, or saturation policy used by the reported object.

A3 is contextual computation rather than a spectral-chart theorem.

All listed artifacts are available in the RIME repository.

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